CS383 · SOFTWARE ENGINEERING

Midterm 451 Solutions

20 marks · four parts A to D

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01 Solution Part A: MCQ

#	Answer	Reason	Lecture
A1	**b**	Microsoft PowerPoint is a generic product because it is sold to any customer on the market. The other two are built for one specific organization, which makes them customized.	1
A2	**b**	One of the SE challenges on the slide: dealing with the high cost of development and maintenance.	1
A3	**c**	The four activities are specification, development (which includes design), validation and evolution. Management is not one of them.	2 · 3
A4	**a**	Team capacity is an input. The sprint goal and the sprint backlog are the **outputs** of sprint planning.	4
A5	**a**	The four factors: process quality, development technology, people quality, and cost, time and schedule.	5
A6	**b**	The design activities: architectural, interface, component and database design.	3

02 Solution Part B: True / False

#	Answer	Reason
B1	**False**	Maintenance and evolution cost more than development, especially for systems with a long life.
B2	**False**	The initial estimate is not accurate. If the estimate is `x`, the actual value ranges from `0.25x` to `4x`, and the estimate improves as the project progresses.

The word always in a true/false statement is usually a sign that it is false.

03 Solution Part C: process model

#	Answer	Reason
C1	**Waterfall**	Safety-critical systems need complete analysis and documentation of all requirements and design before implementation, and each phase is finished and reviewed before the next one. That is the plan-driven style.
C2	**Agile**	Agile delivers small successive versions and the customer is part of the team, so the customer sees the system early and gives feedback on every version.

04 Solution Part D: ethics

D1) Any two examples from the **PUBLIC** principle of the Code, such as:

1. Accept full responsibility for their own work.
2. Approve software only if they have a well-founded belief that it is safe, meets specifications and passes appropriate tests.
3. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment.

Example	Principle	Reason
D2.1) ensure a realistic estimation of cost and plan on all projects	**MANAGEMENT**	Estimating cost and planning are project management work.
D2.2) shouldn't illegally obtain a software application for any purpose	**PROFESSION**	Of the four options this is the closest: obeying the law protects the integrity and reputation of the profession.